

Ideation Phase

Brainstorm & Idea Prioritization

Date	18th June 2026
Student Name	Tirumala Srinidhi
Project Name	ShopEZ – E-Commerce Application Using Mern Stack
Maximum Marks	4 Marks

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Problem Statement:

Online shopping platforms are often complex, confusing, and not user-friendly for both buyers and sellers. Buyers struggle to find products easily, track orders, and complete secure payments. Sellers lack proper tools to manage orders and understand their business performance.

Our Goal:

To build Shop-EZ — a simple, fast, and secure full-stack e-commerce web application where users can browse products, add them to cart, place orders, and track purchases, while admins can manage inventory and monitor sales through a dedicated dashboard.

Step-2: Brainstorm, Idea Listing and Grouping

Group A – User Side Features

- User registration and login with secure JWT authentication
- Browse products with detailed descriptions, images, pricing, and customer reviews
- Add products to cart and manage cart items
- Place orders by entering shipping address and payment details
- View order history and current order status in the profile section
- Role-based access control to separate user and admin permissions.

Group B – Admin Side Features

- Admin login with separate authentication
- Manage product listings — add, update, delete products
- View and manage all customer orders
- Upload and manage banner images and product categories
- Access analytics dashboard to track sales and business performance.

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Group C – Technical Ideas

- MERN Stack (MongoDB, Express.js, React.js, Node.js) for full-stack development
- React.js frontend with component-based UI for fast rendering
- MongoDB collections for Users, Products, Cart, and Orders
- RESTful API design for clean communication between frontend and backend
- Recharts library for visual analytics in the admin dashboard
- JWT tokens for secure session management

Step-3: Idea Prioritization

Idea	Impact (High/Med/Low)	Feasibility (High/Med/Low)	Priority
User Registration & JWT Login	High	High	1st
Product Catalog with Search	High	High	2nd
Add to Cart & Order Placement	High	High	3rd
Secure Checkout with Address & Payment	High	Medium	4th
Order Confirmation & History	High	High	5th
Admin Dashboard with Analytics	High	Medium	6th
Role-Based Access Control	Medium	High	7th
Banner & Category Management	Medium	High	8 th
Recharts Visual Analytics	Medium	Medium	9th

Selected Final Idea:

Shop-EZ — A full-stack MERN e-commerce platform with user-side shopping features (browse, cart, order, track) and admin-side management features (product control, order management, analytics dashboard), built with secure authentication and a clean, responsive interface.